

pyAmpliCol 0.1.0

Methodology and Performance Report

Rikkert Frederix

Valentin Hirschi

Timea Vitos

Release 0.1.0

Abstract

This report defines the public theory, validation, and performance methodology for **pyAmpliCol**, release 0.1.0. It fixes the canonical process coverage for the built-in Standard Model, an externally loaded Standard Model, a scalar contact model, and a scalar-gravity model. Its tables are a reproducible publication surface for measurements produced from the developer installation and recorded with their effective configuration, environment, artifacts, logs, and independent-reference provenance.

Contents

1	Scope	2
2	Current Recursion and Model Lowering	2
2.1	Shared-current DAG	2
2.2	Contact terms and spin two	2
3	Colour and Runtime Semantics	3
4	Benchmark Methodology	3
4.1	Configuration contract	3
4.2	Numerical validation	3
5	Standard-Model Process Matrices	5
5.1	Built-in SM LC process matrix	5
5.2	Built-in SM NLC process matrix	8
5.3	Built-in SM full-colour process matrix	10
5.4	UFO-SM LC process matrix	12
5.5	UFO-SM NLC process matrix	15
5.6	UFO-SM full-colour process matrix	17
6	Z-plus-Jets Evaluator Ladders	19
6.1	Dedicated $d\bar{d} \rightarrow Z + \text{Gluon}$ Performance	20
6.2	UFO-SM Dedicated $d\bar{d} \rightarrow Z + \text{Gluon}$ Performance	21
7	Colour-Singlet Model Ladders	23
7.1	Scalar-contact ladder	23
7.2	Scalar-gravity ladder	23

8 Interpretation

23

1 Scope

pyAmpliCol generates tree-level scattering-amplitude artifacts and evaluates them through the Rusticol runtime. The Python layer resolves models and processes, constructs a shared-current directed acyclic graph (DAG), lowers bounded stages to Symbolica evaluators, and writes a schema-v3 process artifact. The runtime loads that artifact and provides summed and resolved evaluation through one physics contract to Python, Rust 2021, C++17, and Fortran 2008 clients.

The report has three purposes:

1. specify the observables and conditions required for a publishable benchmark;
2. preserve a stable matrix of representative process families and colour approximations; and
3. separate measured data from prose by rendering all result tables from validated JSON caches.

An N/A entry means that the current cache has no recorded measurement for that slot. It is neither a zero, a failure, nor a value carried over from another implementation. Explicit failure labels such as T/O, RAM>800G, VALIDATION FAILED, and ERROR indicate measured campaign outcomes with saved logs.

2 Current Recursion and Model Lowering

2.1 Shared-current DAG

For an external subset I , particle state t , and local spin or flow label χ , an off-shell current has the schematic recursion

$$J_{t,\chi}^\alpha(I) = P^\alpha_\beta(t, p_I) \sum_{I=A \dot{\cup} B} V^\beta_{\gamma\delta}(t_A, t_B \rightarrow t) J_{t_A, \chi_A}^\gamma(A) J_{t_B, \chi_B}^\delta(B), \quad p_I = \sum_{i \in I} p_i. \quad (1)$$

Current identity includes the particle, external labels, helicity ancestry, chirality or spin state, flavour and charge flow, colour state, momentum mask, and any auxiliary-current kind. Contributions with the same complete identity share one node. A topological sweep therefore evaluates each retained current once per phase-space point and reuses it in all dependent amplitudes.

External UFO or serialized JSON sources are compiled to the same canonical model representation as the process generator consumes. Particle metadata, interactions, coupling orders, Lorentz tensors, colour tensors, propagators, and runtime parameters come from that representation. UFO directories contain trusted executable Python; serialized JSON is the portable data-only source.

Optimizations on the external-model path are proof-gated rather than selected by particle names or PDG values. Exact normalized colour and Lorentz expressions, propagator metadata, and coupling relations certify current reflection, helicity, and reusable-kernel identities. For overlapping Standard-Model validation processes, the resulting current, interaction, amplitude-root, and colour-sector counts must match the built-in compatibility model after applying the same independently inferred perturbative-order envelope. An external compiler may retain fewer evaluator calls only when an additional exact kernel-equivalence certificate proves that reuse.

2.2 Contact terms and spin two

A vertex with valence above three is lowered to a deterministic tree of non-propagating auxiliary currents. The construction inserts the original coupling once and introduces no pole. Direct contraction of the original contact tensor is the validation oracle for the scalar-contact ladder.

The scalar-gravity ladder exercises rank-two external states and momentum-dependent interactions. For a massless graviton, the helicity-two polarization tensor is represented as

$$\epsilon_{\pm 2}^{\mu\nu} = \epsilon_\pm^\mu \epsilon_\pm^\nu. \quad (2)$$

Validation checks symmetry, transversality, tracelessness, normalization, and agreement between ordinary and higher-precision evaluation.

3 Colour and Runtime Semantics

The process matrices use three colour contracts:

LC	Leading-colour ordered amplitudes with physical colour-flow selectors.
NLC	The leading term plus the first retained $1/N_c^2$ -suppressed contribution.
Full	The complete supported sparse colour metric for the generated amplitude basis.

The optimized `Runtime.evaluate()` path returns one summed matrix element per point.

The resolved method `Runtime.evaluate_resolved()` exposes LC values with axes `(point, helicity, flow)`. NLC and full-colour artifacts expose one already contracted colour component per helicity. Summing all non-point axes must reproduce the optimized total. Public selectors use physical helicity and flow identifiers; graph-local colour-sector identifiers are not report inputs.

All language bindings load the same process manifest and evaluator payloads. Cross-language agreement therefore validates the runtime interfaces and data layout. A separately generated reference is required for an independent physics comparison.

The default JIT artifact embeds a self-contained SymJIT application for f64 evaluation. Rusticol executes that payload through the separate MIT-licensed SymJIT runtime without importing Symbolica or consulting its runtime license state. Higher-precision Python evaluation and artifacts generated with the ASM or C++ evaluator backends retain an explicit Symbolica runtime capability.

4 Benchmark Methodology

4.1 Configuration contract

Each measurement begins from an immutable resolved configuration. Unless a table variant says otherwise, the report contract uses the release defaults: requested SymJIT optimization level 3, evaluator batch size 128, output chunk size 128, stage-local parameter layout, ten Horner iterations, backend-default common-pair elimination, a ten-second benchmark target, two warmup runs, and at least five timed samples. The process matrices always record requested level 3 in their machine-readable benchmark contract; only explicitly labelled comparison rows, such as JIT level 1 in the dedicated Z ladder, override it. SymJIT may apply its documented backend fallback independently of the requested level. Requested and effective configurations are stored separately so resource or license adjustments remain visible.

Generation time is wall time for one explicit `Generator.generate()` operation starting from the declared model source and ending with a committed, loadable artifact. Model-source conversion, process generation, evaluator compilation, and API-bundle emission must be identified separately when they are measured separately. Existing artifacts may be benchmarked, but their generation slot remains N/A unless a matching generation measurement is available.

Runtime measurements use `BenchmarkRunner` and the summed `Runtime.evaluate()` observable. A cache entry records sample count, wall and evaluator time per point, standard deviation, standard error, relative standard error, effective settings, and environment. Comparisons are valid only for the same process, colour accuracy, parameters, phase-space ensemble, precision, selectors, and normalization.

4.2 Numerical validation

The matrix element and its independent reference are evaluated at the same deterministic points with the same parameter card. Shared-core language checks use relative tolerance 10^{-12} and absolute tolerance 10^{-15} . Independent

reference comparisons use relative tolerance 10^{-8} , with higher precision used to diagnose unstable points. A populated result must also satisfy the resolved-sum identity and preserve expected structural zeros.

No timing is accepted as a numerical validation result. Conversely, a valid matrix element does not populate generation or runtime fields without a matching timing run.

Publication workflow. The report service validates all cache entries, renders every table in a staging directory, optionally compiles the PDF there, and publishes the JSON, TeX, and PDF set only after all steps succeed. Failed refreshes leave the previous published set in place. Physics generation and benchmarking are not implicit report operations.

5 Standard-Model Process Matrices

The canonical matrix spans electroweak vector production, charged-current and leptonic channels, heavy-quark production, pure gluons, multi-boson final states, Higgs-associated final states, and three- and four-quark-line cases. Here n is the number of final-state particles. LC coverage retains the established grid through $n = 9$, with the four bounded families ending at $n = 8$. NLC and full-colour grids retain $n = 1, \dots, 5$; cells below a family's minimum physical multiplicity remain explicitly marked N/A and carry an applicability flag in JSON.

The external-SM matrices use the same process definitions, colour contracts, and numerical input parameters as the built-in matrices. Their model source is the packaged Standard Model loaded through the external JSON/UFO compilation path, so the comparison isolates model-source and generic-lowering behaviour without changing the requested physics or the original Fortran AmpliCol reference.

5.1 Built-in SM LC process matrix

ID base process	n=1					n=2					n=3				
1 $d\bar{d} \rightarrow Z + (n-1)g$	8.19	/0.000392	(x0.0255)	(x247)		0.0581	(N/A)	(N/A)			0.121	(N/A)	(N/A)		
	0.0564	/0.129	(x5.22 1.54)	(x1.24 0.387)		0.00504	(N/A N/A)	(N/A N/A)			0.00542	(N/A N/A)	(N/A N/A)		
2 $u\bar{d} \rightarrow W^+ + (n-1)g$	7.9	/0.000371	(x0.0204)	(x251)		UNSUPPORTED	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
	0.0332	/0.13	(x8.07 2.07)	(x1.22 0.374)		UNSUPPORTED	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
3 $d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						UNSUPPORTED	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
4 $u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						UNSUPPORTED	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
5 $d\bar{d} \rightarrow ZZ + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						UNSUPPORTED	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
6 $gg \rightarrow t\bar{t} + (n-2)g$	N/A					0.0532	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						0.00809	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
7 $d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						UNSUPPORTED	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
8 $gg \rightarrow gg + (n-2)g$	N/A					0.124	(N/A)	(N/A)			UNSUPPORTED	(N/A)	(N/A)		
						0.0049	(N/A N/A)	(N/A N/A)			UNSUPPORTED	(N/A N/A)	(N/A N/A)		
9 $d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A					N/A					UNSUPPORTED	(N/A)	(N/A)		
											UNSUPPORTED	(N/A N/A)	(N/A N/A)		
10 $d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A					N/A					N/A				
11 $d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A					N/A					N/A				
12 $d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A					N/A					N/A				
13 $d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A					N/A					N/A				
14 $d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A					N/A					N/A				
gen O3 one-flow, hel. sum	7.9	8.19	8.04	8.04		0.0532	0.124	0.0784	0.0581		0.121	0.121	0.121	0.121	
	x0.0204	x0.0255	x0.0229	x0.0229	x0.023	x3.04	x7.56	x5.58	x6.15	x4.83	x4.48	x4.48	x4.48	x4.48	x4.48
gen O3 all-flows, one-hel	0.000371	0.000392	0.000382	0.000382		N/A					N/A				
	x247	x251	x249	x249	x249	N/A					N/A				
run O3 one-flow, hel. sum	0.0332	0.0564	0.0448	0.0448		N/A					N/A				
	x5.22	x8.07	x6.65	x6.65	x6.28	N/A					N/A				
run O3 all-flows, one-hel	0.129	0.13	0.13	0.13		N/A					N/A				
	x1.22	x1.24	x1.23	x1.23	x1.23	N/A					N/A				

Built-in SM LC process matrix (continued)

ID	base process	n=4			n=5			n=6
1	$d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	UNSUPPORTED	(N/A N/A)	(N/A N/A)	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	UNSUPPORTED	(N/A N/A)	(N/A N/A)	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
6	$gg \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
8	$gg \rightarrow gg + (n-2)g$	N/A			N/A			N/A
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	UNSUPPORTED	(N/A N/A)	(N/A N/A)	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A			N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)				
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A			N/A			N/A
gen O3 one-flow, hel. sum		N/A			N/A			N/A
		N/A			N/A			N/A
gen O3 all-flows, one-hel		N/A			N/A			N/A
		N/A			N/A			N/A
run O3 one-flow, hel. sum		N/A			N/A			N/A
		N/A			N/A			N/A
run O3 all-flows, one-hel		N/A			N/A			N/A
		N/A			N/A			N/A

Built-in SM LC process matrix (continued)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \rightarrow Z + (n-1)g$	N/A	N/A	N/A
2 $u\bar{d} \rightarrow W^+ + (n-1)g$	N/A	N/A	N/A
3 $d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A	N/A	N/A
4 $u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A	N/A	N/A
5 $d\bar{d} \rightarrow ZZ + (n-2)g$	N/A	N/A	N/A
6 $gg \rightarrow t\bar{t} + (n-2)g$	N/A	N/A	N/A
7 $d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A	N/A	N/A
8 $gg \rightarrow gg + (n-2)g$	N/A	N/A	N/A
9 $d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A	N/A	N/A
10 $d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A	N/A	N/A
11 $d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A	N/A	N/A
12 $d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A	N/A	N/A
13 $d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A	N/A	N/A
14 $d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A	N/A	N/A
gen O3 one-flow, hel. sum	N/A	N/A	N/A
	N/A	N/A	N/A
gen O3 all-flows, one-hel	N/A	N/A	N/A
	N/A	N/A	N/A
run O3 one-flow, hel. sum	N/A	N/A	N/A
	N/A	N/A	N/A
run O3 all-flows, one-hel	N/A	N/A	N/A
	N/A	N/A	N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

5.2 Built-in SM NLC process matrix

ID	base process	n=1				n=2				n=3			
1	$d\bar{d} \rightarrow Z + (n-1)g$	0.0587 s	(x5.34)			0.053 s	(x6.15)			0.129 s	(x4.16)		
		5.86e3 us	(x0.000187 0.000187)			6.02e3 us	(x0.000287 0.000287)			5.56e3 us	(x0.000742 0.000742)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	0.0509 s	(x7.25)			UNSUPPORTED	0.329 s			UNSUPPORTED	0.459 s		
		4.53e3 us	(x0.000229 0.000229)			UNSUPPORTED	1.75 us 1.75 us			UNSUPPORTED	3.42 us 3.42 us		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A				UNSUPPORTED	0.345 s			UNSUPPORTED	0.321 s		
						UNSUPPORTED	1.29 us 1.29 us			UNSUPPORTED	1.8 us 1.8 us		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A				UNSUPPORTED	0.397 s			UNSUPPORTED	0.292 s		
						UNSUPPORTED	1.19 us 1.19 us			UNSUPPORTED	1.6 us 1.6 us		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	N/A				UNSUPPORTED	0.284 s			UNSUPPORTED	0.505 s		
						UNSUPPORTED	1.75 us 1.75 us			UNSUPPORTED	3.11 us 3.11 us		
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A				0.0569 s	(x5.8)			UNSUPPORTED	1.51 s		
						5.05e3 us	(x0.000506 0.000506)			UNSUPPORTED	23.9 us 23.9 us		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A				UNSUPPORTED	0.282 s			UNSUPPORTED	0.71 s		
						UNSUPPORTED	2.12 us 2.12 us			UNSUPPORTED	9.71 us 9.71 us		
8	$gg \rightarrow gg + (n-2)g$	N/A				0.13 s	(x4.44)			UNSUPPORTED	4.99 s		
						5.24e3 us	(x0.000661 0.000661)			UNSUPPORTED	80.3 us 80.3 us		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A				N/A				UNSUPPORTED	0.77 s		
										UNSUPPORTED	5.04 us 5.04 us		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A				N/A				N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A				N/A				N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A				N/A				N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A				N/A				N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A				N/A				N/A			
summary: gen		0.0509	0.0587	0.0548	0.0548	0.053	0.13	0.0799	0.0569	0.129	0.129	0.129	0.129
		(x5.34)	(x7.25)	(x6.29)	(x6.29)	(x4.44)	(x6.15)	(x5.46)	(x5.8)	(x4.16)	(x4.16)	(x4.16)	(x4.16)
summary: run wall		4.53e3	5.86e3	5.19e3	5.19e3	5.05e3	6.02e3	5.44e3	5.24e3	5.56e3	5.56e3	5.56e3	5.56e3
		(x0.000187	x0.000229	x0.000206	x0.000205)	(x0.000287	x0.000661	x0.000486	x0.000506	(x0.000742	x0.000742	x0.000742	x0.000742)
summary: run eval		4.53e3	5.86e3	5.19e3	5.19e3	5.05e3	6.02e3	5.44e3	5.24e3	5.56e3	5.56e3	5.56e3	5.56e3
		(x0.000187	x0.000229	x0.000206	x0.000205)	(x0.000287	x0.000661	x0.000486	x0.000506	(x0.000742	x0.000742	x0.000742	x0.000742)

Built-in SM NLC process matrix (continued)

ID base process	n=4	n=5
1 $d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED UNSUPPORTED	3.85 s 53.1 us 53.1 us
2 $u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED UNSUPPORTED	2.74 s 46.1 us 46.1 us
3 $d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED UNSUPPORTED	0.605 s 4.14 us 4.14 us
4 $u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED UNSUPPORTED	0.558 s 3.15 us 3.15 us
5 $d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED UNSUPPORTED	1.79 s 17.8 us 17.8 us
6 $gg \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED UNSUPPORTED	21.6 s 736 us 736 us
7 $d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED UNSUPPORTED	5.92 s 169 us 169 us
8 $gg \rightarrow gg + (n-2)g$	N/A	N/A
9 $d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED UNSUPPORTED	1.77 s 14 us 14 us
10 $d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.747 s 4.76 us 4.76 us
11 $d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED UNSUPPORTED	1.36 s 18.3 us 18.3 us
12 $d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.592 s 4.17 us 4.17 us
13 $d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.798 s 11.1 us 11.1 us
14 $d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A	N/A
summary: gen	N/A N/A	N/A N/A
summary: run wall	N/A N/A	N/A N/A
summary: run eval	N/A N/A	N/A N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

5.3 Built-in SM full-colour process matrix

ID	base process	n=1					n=2				n=3			
1	$d\bar{d} \rightarrow Z + (n-1)g$	0.0497 s	(x6.56)				0.0539 s	(x5.16)			0.124 s	(x4.69)		
		5.29e3 us	(x0.000213 0.000213)				8.01e3 us	(x0.000202 0.000202)			5.86e3 us	(x0.000721 0.000721)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	0.0608 s	(x4.42)				UNSUPPORTED	0.274 s			UNSUPPORTED	0.473 s		
		6.12e3 us	(x0.000194 0.000194)				UNSUPPORTED	1.51 us 1.51 us			UNSUPPORTED	3.47 us 3.47 us		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A					UNSUPPORTED	0.258 s			UNSUPPORTED	0.298 s		
							UNSUPPORTED	1.24 us 1.24 us			UNSUPPORTED	1.8 us 1.8 us		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A					UNSUPPORTED	0.244 s			UNSUPPORTED	0.283 s		
							UNSUPPORTED	1.16 us 1.16 us			UNSUPPORTED	1.6 us 1.6 us		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	N/A					UNSUPPORTED	0.289 s			UNSUPPORTED	0.489 s		
							UNSUPPORTED	1.75 us 1.75 us			UNSUPPORTED	3.16 us 3.16 us		
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A					0.0511 s	(x6.96)			UNSUPPORTED	1.69 s		
							4.99e3 us	(x0.000514 0.000514)			UNSUPPORTED	29.6 us 29.6 us		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A					UNSUPPORTED	0.29 s			UNSUPPORTED	0.849 s		
							UNSUPPORTED	2.12 us 2.12 us			UNSUPPORTED	13.8 us 13.8 us		
8	$gg \rightarrow gg + (n-2)g$	N/A					0.119 s	(x3.9)			UNSUPPORTED	5.13 s		
							5.65e3 us	(x0.000616 0.000616)			UNSUPPORTED	93.1 us 93.1 us		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A					N/A				UNSUPPORTED	0.803 s		
											UNSUPPORTED	5.55 us 5.55 us		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A					N/A				N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A					N/A				N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A					N/A				N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A					N/A				N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A					N/A				N/A			
summary: gen		0.0497	0.0608	0.0553	0.0553		0.0511	0.119	0.0746	0.0539	0.124	0.124	0.124	0.124
		(x4.42)	(x6.56)	(x5.49)	(x5.49)	(x5.38)	(x3.9)	(x6.96)	(x5.34)	(x5.16)	(x4.69)	(x4.69)	(x4.69)	(x4.69)
summary: run wall		5.29e3	6.12e3	5.7e3	5.7e3		4.99e3	8.01e3	6.21e3	5.65e3	5.86e3	5.86e3	5.86e3	5.86e3
		(x0.000194	x0.000213	x0.000203	x0.000203	x0.000203)	(x0.000202	x0.000616	x0.000446	x0.000516	(x0.000721	x0.000721	x0.000721	x0.000721)
summary: run eval		5.29e3	6.12e3	5.7e3	5.7e3		4.99e3	8.01e3	6.21e3	5.65e3	5.86e3	5.86e3	5.86e3	5.86e3
		(x0.000194	x0.000213	x0.000203	x0.000203	x0.000203)	(x0.000202	x0.000616	x0.000446	x0.000516	(x0.000721	x0.000721	x0.000721	x0.000721)

Built-in SM full-colour process matrix (continued)

ID base process	n=4	n=5
1 $d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED UNSUPPORTED	3.65 s 52.6 us 52.6 us
2 $u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED UNSUPPORTED	2.27 s 41.4 us 41.4 us
3 $d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED UNSUPPORTED	0.594 s 4.08 us 4.08 us
4 $u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED UNSUPPORTED	0.407 s 3.11 us 3.11 us
5 $d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED UNSUPPORTED	1.71 s 22 us 22 us
6 $gg \rightarrow t\bar{t} + (n-2)g$	N/A	N/A
7 $d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED UNSUPPORTED	5.69 s 168 us 168 us
8 $gg \rightarrow gg + (n-2)g$	N/A	N/A
9 $d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED UNSUPPORTED	1.79 s 17.8 us 17.8 us
10 $d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.647 s 5.06 us 5.06 us
11 $d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED UNSUPPORTED	1.43 s 16.5 us 16.5 us
12 $d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.612 s 4.09 us 4.09 us
13 $d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED UNSUPPORTED	0.811 s 11.3 us 11.3 us
14 $d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A	N/A
summary: gen	N/A N/A	N/A N/A
summary: run wall	N/A N/A	N/A N/A
summary: run eval	N/A N/A	N/A N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

5.4 UFO-SM LC process matrix

ID	base process	n=1					n=2				n=3			
1	$d\bar{d} \rightarrow Z + (n-1)g$	8.46	/ 0.00082	(x0.0319)	(x246)		0.0633	(N/A)	(N/A)		0.141	(N/A)	(N/A)	
		0.055	/ 0.136	(x5.59 1.69)	(x1.19 0.369)		0.00594	(N/A N/A)	(N/A N/A)		0.00637	(N/A N/A)	(N/A N/A)	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	8.52	/0.00037	(x0.0304)	(x526)		UNSUPPORTED	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
		0.0331	/0.131	(x8.2 1.98)	(x1.25 0.391)		UNSUPPORTED	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							UNSUPPORTED	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							UNSUPPORTED	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							UNSUPPORTED	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	N/A					0.0506	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							0.00597	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A					UNSUPPORTED	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							UNSUPPORTED	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	N/A					0.118	(N/A)	(N/A)		UNSUPPORTED	(N/A)	(N/A)	
							0.0052	(N/A N/A)	(N/A N/A)		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A					N/A				UNSUPPORTED	(N/A)	(N/A)	
											UNSUPPORTED	(N/A N/A)	(N/A N/A)	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A					N/A				N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A					N/A				N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A					N/A				N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A					N/A				N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A					N/A				N/A			
gen O3 one-flow, hel. sum		8.46	8.52	8.49	8.49		0.0506	0.118	0.0772	0.0633		0.141	0.141	0.141
		x0.0304	x0.0319	x0.0312	x0.0312	x0.0312	x3.79	x9.15	x6.42	x6.31	x5.65	x4.94	x4.94	x4.94
gen O3 all-flows, one-hel		0.00037	0.00082	0.000595	0.000595		N/A					N/A		
		x246	x526	x386	x386	x333	N/A					N/A		
run O3 one-flow, hel. sum		0.0331	0.055	0.044	0.044		N/A					N/A		
		x5.59	x8.2	x6.89	x6.89	x6.57	N/A					N/A		
run O3 all-flows, one-hel		0.131	0.136	0.133	0.133		N/A					N/A		
		x1.19	x1.25	x1.22	x1.22	x1.22	N/A					N/A		

UFO-SM LC process matrix (continued)

ID	base process	n=4		n=5	n=6
1	$d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
6	$gg \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
8	$gg \rightarrow gg + (n-2)g$	N/A		N/A	N/A
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED	(N/A)	(N/A)	N/A
		UNSUPPORTED	(N/A N/A)	(N/A N/A)	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A		N/A	N/A
gen O3 one-flow, hel. sum		N/A		N/A	N/A
		N/A		N/A	N/A
gen O3 all-flows, one-hel		N/A		N/A	N/A
		N/A		N/A	N/A
run O3 one-flow, hel. sum		N/A		N/A	N/A
		N/A		N/A	N/A
run O3 all-flows, one-hel		N/A		N/A	N/A
		N/A		N/A	N/A

UFO-SM LC process matrix (continued)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \rightarrow Z + (n-1)g$	N/A	N/A	N/A
2 $u\bar{d} \rightarrow W^+ + (n-1)g$	N/A	N/A	N/A
3 $d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A	N/A	N/A
4 $u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A	N/A	N/A
5 $d\bar{d} \rightarrow ZZ + (n-2)g$	N/A	N/A	N/A
6 $gg \rightarrow t\bar{t} + (n-2)g$	N/A	N/A	N/A
7 $d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A	N/A	N/A
8 $gg \rightarrow gg + (n-2)g$	N/A	N/A	N/A
9 $d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A	N/A	N/A
10 $d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A	N/A	N/A
11 $d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A	N/A	N/A
12 $d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A	N/A	N/A
13 $d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A	N/A	N/A
14 $d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A	N/A	N/A
gen O3 one-flow, hel. sum	N/A	N/A	N/A
	N/A	N/A	N/A
gen O3 all-flows, one-hel	N/A	N/A	N/A
	N/A	N/A	N/A
run O3 one-flow, hel. sum	N/A	N/A	N/A
	N/A	N/A	N/A
run O3 all-flows, one-hel	N/A	N/A	N/A
	N/A	N/A	N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

5.5 UFO-SM NLC process matrix

ID	base process	n=1					n=2				n=3			
1	$d\bar{d} \rightarrow Z + (n-1)g$	0.053 s	(x8.29)				0.074 s	(x5.43)			0.147 s	(x4.79)		
		4.44e3 us	(x0.00026 0.00026)				4.81e3 us	(x0.000335 0.000335)			6.36e3 us	(x0.000728 0.000728)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	0.0608 s	(x6.96)				UNSUPPORTED	0.398 s			UNSUPPORTED	0.611 s		
		5.86e3 us	(x0.000179 0.000179)				UNSUPPORTED	1.5 us 1.5 us			UNSUPPORTED	3.57 us 3.57 us		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A					UNSUPPORTED	0.398 s			UNSUPPORTED	0.501 s		
							UNSUPPORTED	1.26 us 1.26 us			UNSUPPORTED	1.89 us 1.89 us		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A					UNSUPPORTED	0.38 s			UNSUPPORTED	0.452 s		
							UNSUPPORTED	1.15 us 1.15 us			UNSUPPORTED	1.79 us 1.79 us		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	N/A					UNSUPPORTED	0.418 s			UNSUPPORTED	0.584 s		
							UNSUPPORTED	1.78 us 1.78 us			UNSUPPORTED	3.31 us 3.31 us		
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A					0.0581 s	(x8.1)			UNSUPPORTED	2 s		
							5.17e3 us	(x0.000497 0.000497)			UNSUPPORTED	27.4 us 27.4 us		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A					UNSUPPORTED	0.464 s			UNSUPPORTED	0.937 s		
							UNSUPPORTED	2.15 us 2.15 us			UNSUPPORTED	8.22 us 8.22 us		
8	$gg \rightarrow gg + (n-2)g$	N/A					0.119 s	(x5.09)			UNSUPPORTED	6.09 s		
							6.19e3 us	(x0.000581 0.000581)			UNSUPPORTED	91.4 us 91.4 us		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A					N/A				UNSUPPORTED	1.12 s		
											UNSUPPORTED	5.8 us 5.8 us		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A					N/A				N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A					N/A				N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A					N/A				N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A					N/A				N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A					N/A				N/A			
summary: gen		0.053	0.0608	0.0569	0.0569		0.0581	0.119	0.0837	0.074	0.147	0.147	0.147	0.147
		(x6.96)	(x8.29)	(x7.63)	(x7.63)	(x7.58)	(x5.09)	(x8.1)	(x6.2)	(x5.43)	(x4.79)	(x4.79)	(x4.79)	(x4.79)
summary: run wall		4.44e3	5.86e3	5.15e3	5.15e3		4.81e3	6.19e3	5.39e3	5.17e3	6.36e3	6.36e3	6.36e3	6.36e3
		(x0.000179	x0.00026	x0.00022	x0.00022	x0.000214)	(x0.000335	x0.000581	x0.000476	x0.000497	(x0.000728	x0.000728	x0.000728	x0.000728)
summary: run eval		4.44e3	5.86e3	5.15e3	5.15e3		4.81e3	6.19e3	5.39e3	5.17e3	6.36e3	6.36e3	6.36e3	6.36e3
		(x0.000179	x0.00026	x0.00022	x0.00022	x0.000214)	(x0.000335	x0.000581	x0.000476	x0.000497	(x0.000728	x0.000728	x0.000728	x0.000728)

UFO-SM NLC process matrix (continued)

ID	base process	n=4	n=5	
1	$d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED	4.43 s	N/A
		UNSUPPORTED	55.6 us 55.6 us	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED	2.98 s	N/A
		UNSUPPORTED	39.1 us 39.1 us	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED	0.836 s	N/A
		UNSUPPORTED	5.28 us 5.28 us	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED	0.659 s	N/A
		UNSUPPORTED	3.07 us 3.07 us	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED	2.19 s	N/A
		UNSUPPORTED	25.9 us 25.9 us	
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A		N/A
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	8.26 s	N/A
		UNSUPPORTED	132 us 132 us	
8	$gg \rightarrow gg + (n-2)g$	N/A		N/A
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED	2.16 s	N/A
		UNSUPPORTED	17.1 us 17.1 us	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED	1.11 s	N/A
		UNSUPPORTED	4.66 us 4.66 us	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED	1.95 s	N/A
		UNSUPPORTED	16.4 us 16.4 us	
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED	0.967 s	N/A
		UNSUPPORTED	4 us 4 us	
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED	1.04 s	N/A
		UNSUPPORTED	8.94 us 8.94 us	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A		N/A
summary: gen		N/A		N/A
		N/A		N/A
summary: run wall		N/A		N/A
		N/A		N/A
summary: run eval		N/A		N/A
		N/A		N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

5.6 UFO-SM full-colour process matrix

ID	base process	n=1					n=2				n=3			
1	$d\bar{d} \rightarrow Z + (n-1)g$	0.0502 s	(x9.02)				0.0614 s	(x7.35)			0.141 s	(x5.12)		
		4.82e3 us	(x0.000227 0.000227)				5.32e3 us	(x0.000313 0.000313)			6.76e3 us	(x0.00065 0.00065)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	0.0567 s	(x8.34)				UNSUPPORTED	0.404 s			UNSUPPORTED	0.593 s		
		7.33e3 us	(x0.000148 0.000148)				UNSUPPORTED	1.53 us 1.53 us			UNSUPPORTED	3.55 us 3.55 us		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	N/A					UNSUPPORTED	0.481 s			UNSUPPORTED	0.464 s		
							UNSUPPORTED	1.27 us 1.27 us			UNSUPPORTED	1.87 us 1.87 us		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	N/A					UNSUPPORTED	0.386 s			UNSUPPORTED	0.415 s		
							UNSUPPORTED	1.21 us 1.21 us			UNSUPPORTED	1.77 us 1.77 us		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	N/A					UNSUPPORTED	0.456 s			UNSUPPORTED	0.581 s		
							UNSUPPORTED	1.86 us 1.86 us			UNSUPPORTED	3.3 us 3.3 us		
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A					0.0582 s	(x9.14)			UNSUPPORTED	2.05 s		
							5.07e3 us	(x0.000515 0.000515)			UNSUPPORTED	29.3 us 29.3 us		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	N/A					UNSUPPORTED	0.437 s			UNSUPPORTED	0.978 s		
							UNSUPPORTED	2.19 us 2.19 us			UNSUPPORTED	10.4 us 10.4 us		
8	$gg \rightarrow gg + (n-2)g$	N/A					0.119 s	(x5.48)			UNSUPPORTED	6.47 s		
							4.84e3 us	(x0.000765 0.000765)			UNSUPPORTED	107 us 107 us		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	N/A					N/A				UNSUPPORTED	0.927 s		
											UNSUPPORTED	7.31 us 7.31 us		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	N/A					N/A				N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	N/A					N/A				N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	N/A					N/A				N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	N/A					N/A				N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A					N/A				N/A			
summary: gen		0.0502	0.0567	0.0534	0.0534		0.0582	0.119	0.0795	0.0614	0.141	0.141	0.141	0.141
		(x8.34)	(x9.02)	(x8.68)	(x8.68)	(x8.66)	(x5.48)	(x9.14)	(x7.33)	(x7.35)	(x5.12)	(x5.12)	(x5.12)	(x5.12)
summary: run wall		4.82e3	7.33e3	6.07e3	6.07e3		4.84e3	5.32e3	5.08e3	5.07e3	6.76e3	6.76e3	6.76e3	6.76e3
		(x0.000148	x0.000227	x0.000180	x0.000180	x0.00018)	(x0.000313	x0.000765	x0.000536	x0.000515	(x0.00065	x0.00065	x0.00065	x0.00065)
summary: run eval		4.82e3	7.33e3	6.07e3	6.07e3		4.84e3	5.32e3	5.08e3	5.07e3	6.76e3	6.76e3	6.76e3	6.76e3
		(x0.000148	x0.000227	x0.000180	x0.000180	x0.00018)	(x0.000313	x0.000765	x0.000536	x0.000515	(x0.00065	x0.00065	x0.00065	x0.00065)

UFO-SM full-colour process matrix (continued)

ID	base process	n=4		n=5
1	$d\bar{d} \rightarrow Z + (n-1)g$	UNSUPPORTED	3.83 s	N/A
		UNSUPPORTED	54.8 us 54.8 us	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	UNSUPPORTED	2.95 s	N/A
		UNSUPPORTED	37.9 us 37.9 us	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	UNSUPPORTED	0.71 s	N/A
		UNSUPPORTED	4.76 us 4.76 us	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	UNSUPPORTED	0.639 s	N/A
		UNSUPPORTED	3.06 us 3.06 us	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	UNSUPPORTED	2.06 s	N/A
		UNSUPPORTED	18.9 us 18.9 us	
6	$gg \rightarrow t\bar{t} + (n-2)g$	N/A		N/A
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	UNSUPPORTED	7.03 s	N/A
		UNSUPPORTED	118 us 118 us	
8	$gg \rightarrow gg + (n-2)g$	N/A		N/A
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	UNSUPPORTED	2.34 s	N/A
		UNSUPPORTED	17.7 us 17.7 us	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	UNSUPPORTED	0.889 s	N/A
		UNSUPPORTED	4.21 us 4.21 us	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	UNSUPPORTED	1.93 s	N/A
		UNSUPPORTED	14.4 us 14.4 us	
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	UNSUPPORTED	0.954 s	N/A
		UNSUPPORTED	4.02 us 4.02 us	
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	UNSUPPORTED	1.23 s	N/A
		UNSUPPORTED	10.3 us 10.3 us	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	N/A		N/A
summary: gen		N/A		N/A
		N/A		N/A
summary: run wall		N/A		N/A
		N/A		N/A
summary: run eval		N/A		N/A
		N/A		N/A

Matrix cells use the old matrix layout: the upper line shows original Fortran AmpliCol generation time and the pyAmpliCol JIT O3 generation ratio; the lower line shows original Fortran AmpliCol runtime and the pyAmpliCol JIT O3 wall|evaluator runtime ratios. Ratios are pyAmpliCol divided by AmpliCol, so values below one mean pyAmpliCol is faster.

6 Z-plus-Jets Evaluator Ladders

The dedicated $d\bar{d} \rightarrow Z + (n-1)g$ ladders compare the independent reference with JIT level 1, JIT level 3, ASM O3, and C++ O3 evaluator configurations. They retain final-state multiplicities $n = 1, \dots, 9$ for the built-in model and $n = 1, \dots, 6$ for the external-model campaign. Variant overrides are stored as typed dotted configuration paths in the cache rather than inferred from table labels.

6.1 Dedicated $d\bar{d} \rightarrow Z + \text{Gluon}$ Performance

n	process	route	setup	selected flow, helicity sum			all flows, fixed helicity		
				gen [s]	wall [us/pt]	eval [us/pt]	gen [s]	wall [us/pt]	eval [us/pt]
1	$d\bar{d}^* \rightarrow Z + (1-1)^*g$	reference	AmpliCol	7.95	0.0546	N/A	0.000817	0.134	N/A
1	$d\bar{d}^* \rightarrow Z + (1-1)^*g$	Rusticol	pyAmpliCol JIT O1	0.236 (x0.03)	1.08 (x19.8)	1.08 (x19.8)	0.0925 (x113)	0.925 (x6.91)	0.925 (x6.91)
1	$d\bar{d}^* \rightarrow Z + (1-1)^*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	UNSUPPORTED	N/A	N/A
1	$d\bar{d}^* \rightarrow Z + (1-1)^*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	UNSUPPORTED	N/A	N/A
1	$d\bar{d}^* \rightarrow Z + (1-1)^*g$	Rusticol	pyAmpliCol JIT O3	0.24 (x0.03)	1.08 (x19.7)	1.08 (x19.7)	0.0895 (x110)	0.93 (x6.94)	0.93 (x6.94)
2	$d\bar{d}^* \rightarrow Z + (2-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
2	$d\bar{d}^* \rightarrow Z + (2-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
2	$d\bar{d}^* \rightarrow Z + (2-1)^*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A
2	$d\bar{d}^* \rightarrow Z + (2-1)^*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A
2	$d\bar{d}^* \rightarrow Z + (2-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d}^* \rightarrow Z + (3-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d}^* \rightarrow Z + (3-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d}^* \rightarrow Z + (3-1)^*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d}^* \rightarrow Z + (3-1)^*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d}^* \rightarrow Z + (3-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
4	$d\bar{d}^* \rightarrow Z + (4-1)^*g$	reference	AmpliCol	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A
4	$d\bar{d}^* \rightarrow Z + (4-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
4	$d\bar{d}^* \rightarrow Z + (4-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
4	$d\bar{d}^* \rightarrow Z + (4-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
4	$d\bar{d}^* \rightarrow Z + (4-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d}^* \rightarrow Z + (5-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d}^* \rightarrow Z + (5-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d}^* \rightarrow Z + (5-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d}^* \rightarrow Z + (5-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d}^* \rightarrow Z + (5-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
6	$d\bar{d}^* \rightarrow Z + (6-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
6	$d\bar{d}^* \rightarrow Z + (6-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
6	$d\bar{d}^* \rightarrow Z + (6-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
6	$d\bar{d}^* \rightarrow Z + (6-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
6	$d\bar{d}^* \rightarrow Z + (6-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d}^* \rightarrow Z + (7-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d}^* \rightarrow Z + (7-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d}^* \rightarrow Z + (7-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d}^* \rightarrow Z + (7-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d}^* \rightarrow Z + (7-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
8	$d\bar{d}^* \rightarrow Z + (8-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
8	$d\bar{d}^* \rightarrow Z + (8-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8	$d\bar{d}^* \rightarrow Z + (8-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
8	$d\bar{d}^* \rightarrow Z + (8-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
8	$d\bar{d}^* \rightarrow Z + (8-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d}^* \rightarrow Z + (9-1)^*g$	reference	AmpliCol	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d}^* \rightarrow Z + (9-1)^*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d}^* \rightarrow Z + (9-1)^*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d}^* \rightarrow Z + (9-1)^*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d}^* \rightarrow Z + (9-1)^*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A

Reproduce the $n = 1$ JIT O1 selected-flow entry with the following generation and timing commands.

```
.venv/bin/python -m pyamplicol generate-process 'd d* > z' .artifacts/performance-report/manual/z_builtin_sm/n1/jit_o1 --replace --color-accuracy lc --batch-size 64 --symbolica-output-chunk-size 128 --reference-color-order 2,1,3 --lc-sector-ids 0
--symbolica-evaluator-backend jit --symbolica-jit-optimization-level 1
.venv/bin/python -m pyamplicol time-process --target-runtime 10 --batch-size 64 --json .artifacts/performance-report/manual/z_builtin_sm/n1/jit_o1
```

pyAmpliCol model source: **Built-in Standard Model**. Here n is final-state multiplicity. Each block reports generation seconds and wall/evaluator microseconds per point: one selected flow with the helicity sum on the left, all flows at one fixed helicity on the right. Wall measures Rusticol's public `evaluate` call; eval separately profiles the evaluator when available. Both use batch 64, stage-local inputs, ten Horner iterations, and a ten-second timing target; output chunks are 128 selected and 8192 all-flow. AmpliCol's generated-library time is reported under wall; its separate eval metric is N/A.

6.2 UFO-SM Dedicated $d\bar{d} \rightarrow Z + \text{Gluon}$ Performance

n	process	route	setup	selected flow, helicity sum				all flows, fixed helicity					
				gen [s]	vs blt-in	wall [us/pt]	vs blt-in	eval [us/pt]	gen [s]	vs blt-in	wall [us/pt]	vs blt-in	eval [us/pt]
1	$d^- \rightarrow Z + (1-1)*g$	reference	AmpliCol	N/A		N/A		N/A	N/A		N/A		N/A
1	$d^- \rightarrow Z + (1-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1	$d^- \rightarrow Z + (1-1)*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1	$d^- \rightarrow Z + (1-1)*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1	$d^- \rightarrow Z + (1-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$d^- \rightarrow Z + (2-1)*g$	reference	AmpliCol	N/A		N/A		N/A	N/A		N/A		N/A
2	$d^- \rightarrow Z + (2-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$d^- \rightarrow Z + (2-1)*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$d^- \rightarrow Z + (2-1)*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$d^- \rightarrow Z + (2-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d^- \rightarrow Z + (3-1)*g$	reference	AmpliCol	N/A		N/A		N/A	N/A		N/A		N/A
3	$d^- \rightarrow Z + (3-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d^- \rightarrow Z + (3-1)*g$	Rusticol	pyAmpliCol ASM O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d^- \rightarrow Z + (3-1)*g$	Rusticol	pyAmpliCol C++ O3	UNSUPPORTED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d^- \rightarrow Z + (3-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	$d^- \rightarrow Z + (4-1)*g$	reference	AmpliCol	UNSUPPORTED		N/A		N/A	N/A		N/A		N/A
4	$d^- \rightarrow Z + (4-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	$d^- \rightarrow Z + (4-1)*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	$d^- \rightarrow Z + (4-1)*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	$d^- \rightarrow Z + (4-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d^- \rightarrow Z + (5-1)*g$	reference	AmpliCol	N/A		N/A		N/A	N/A		N/A		N/A
5	$d^- \rightarrow Z + (5-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d^- \rightarrow Z + (5-1)*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d^- \rightarrow Z + (5-1)*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d^- \rightarrow Z + (5-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$d^- \rightarrow Z + (6-1)*g$	reference	AmpliCol	N/A		N/A		N/A	N/A		N/A		N/A
6	$d^- \rightarrow Z + (6-1)*g$	Rusticol	pyAmpliCol JIT O1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$d^- \rightarrow Z + (6-1)*g$	Rusticol	pyAmpliCol ASM O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$d^- \rightarrow Z + (6-1)*g$	Rusticol	pyAmpliCol C++ O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$d^- \rightarrow Z + (6-1)*g$	Rusticol	pyAmpliCol JIT O3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

pyAmpliCol model source: UFO-SM. Here n is final-state multiplicity. Each block reports generation seconds and wall/evaluator microseconds per point: one selected flow with the helicity sum on the left, all flows at one fixed helicity on the right. Wall measures Rusticol's public `evaluate` call; eval separately profiles the evaluator when available. Both use batch 64, stage-local inputs, ten Horner iterations, and a ten-second timing target; output chunks are 128 selected and 8192 all-flow. AmpliCol's generated-library time is reported under wall; its separate eval metric is N/A. In this UFO-SM table, each `vs blt-in` entry is the adjacent generation- or wall-time ratio to the matching built-in-SM row at the same multiplicity, backend, and flow/helicity workload; values below one mean that UFO-SM is faster.

Z tables use the old performance-summary layout: reference rows are original Fortran AmpliCol, pyAmpliCol rows show generation, wall, and evaluator timings with ratios against the reference where available. Variant overrides are machine-readable in JSON.

7 Colour-Singlet Model Ladders

The scalar-contact ladder retains $n = 2, \dots, 8$ final scalars. It tests generic high-valence contact decomposition, identical-particle normalization, and source parity without Standard-Model-specific kernels. The scalar-gravity ladder retains $n = 2, \dots, 4$ final gravitons and exercises spin-two wavefunctions, tensor propagators, and momentum-dependent vertices. Since all external particles in these fixtures are colour singlets, LC, NLC, and full colour reduce to the same unit colour contraction.

7.1 Scalar-contact ladder

metric	scalar_0	scalar_0 > X*scalar_0					
	$X = 2$	$X = 3$	$X = 4$	$X = 5$	$X = 6$	$X = 7$	$X = 8$
status	OK	OK	OK	N/A	N/A	N/A	N/A
generation [s]	1.43	1.90	4.34	N/A	N/A	N/A	N/A
wall [s/pt]	1.52×10^{-6}	2.15×10^{-6}	3.13×10^{-6}	N/A	N/A	N/A	N/A
evaluator [s/pt]	N/A	N/A	N/A	N/A	N/A	N/A	N/A
matrix element	5.00×10^{-1}	1.67×10^{-1}	4.17×10^{-2}	N/A	N/A	N/A	N/A
relative difference	1.11×10^{-16}	1.67×10^{-16}	0	N/A	N/A	N/A	N/A

7.2 Scalar-gravity ladder

metric	scalar_0	scalar_0 > X*graviton		
	$X = 2$	$X = 3$	$X = 4$	
status	OK	N/A	N/A	
generation [s]	2.81	N/A	N/A	
wall [s/pt]	4.64×10^{-6}	N/A	N/A	
evaluator [s/pt]	N/A	N/A	N/A	
matrix element	1.61×10^9	N/A	N/A	
relative difference	1.04×10^{-15}	N/A	N/A	

8 Interpretation

This report establishes coverage, methodology, and the current measured cache state. Performance regression gates should be enabled only after the same developer-installed artifact has produced a complete, reviewed baseline on named hardware. Consumers should treat every N/A entry as unavailable and should not interpolate, copy, or estimate a value from neighbouring multiplicities or model profiles.

The machine-readable caches under `docs/results/` are authoritative for table completeness and measurement provenance. The TeX inputs and this PDF are derived publication artifacts.